

1 状态变量、输入变量，输出变量

状态变量7个

- 1 'Uc_Cp'
- 2 'Uc_Cps'
- 3 'Uc-Cs'
- 4 'Il_Lph1'
- 5 'Il_Lph2'
- 6 'Il_Ls'
- 7 'Il_Lm: Linear Transformer'

输入变量18个

- 1 'LS1/Ideal Switch'
- 2 'LS2/Ideal Switch'
- 3 'LS3/Ideal Switch'
- 4 'LS4/Ideal Switch'
- 5 'LS5/Ideal Switch'
- 6 'LS6/Ideal Switch'
- 7 'LS7/Ideal Switch'
- 8 'LS8/Ideal Switch'
- 9 'LS1/Diode'
- 10 'LS2/Diode'
- 11 'LS3/Diode'
- 12 'LS4/Diode'
- 13 'LS5/Diode'
- 14 'LS6/Diode'
- 15 'LS7/Diode'
- 16 'LS8/Diode'
- 17 'U_Udc1'
- 18 'U_Udc2'

输出变量18个

- 1 'U_S1/Ideal Switch'
- 2 'U_S2/Ideal Switch'
- 3 'U_S3/Ideal Switch'
- 4 'U_S4/Ideal Switch'
- 5 'U_S5/Ideal Switch'

- 6 'U_S6/Ideal Switch'
- 7 'U_S7/Ideal Switch'
- 8 'U_S8/Ideal Switch'
- 9 'U_S1/Diode'
- 10 'U_S2/Diode'
- 11 'U_S3/Diode'
- 12 'U_S4/Diode'
- 13 'U_S5/Diode'
- 14 'U_S6/Diode'
- 15 'U_S7/Diode'
- 16 'U_S8/Diode'
- 17 'U_Voltage Measurement'
- 18 'U_Voltage Measurement1'
- 19 'I_Current Measurement'

2 simulink建模

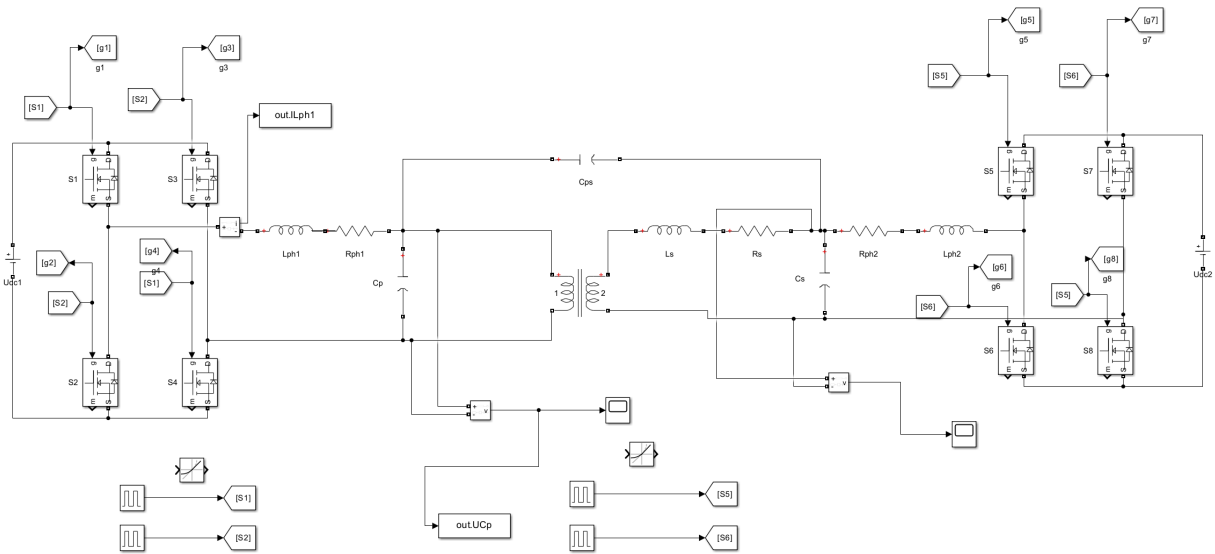


图 1: simmulink模型

3 控制信号

S1和S4控制信号相同:

$g1=g4$ (方波, 周期 $100\text{e-}6$ 秒, 占空比50%, 相位延迟0秒)

S2和S3控制信号相同(与S1、S4互补):

$g2=g3$ (方波, 周期 $100\text{e-}6$ 秒, 占空比50%, 相位延迟 $50\text{e-}6$ 秒)

S5和S8控制信号相同(延迟S1、S4四分之一周期):

$g5=g8$ (方波, 周期 $100\text{e-}6$ 秒, 占空比50%, 相位延迟 $25\text{e-}6$ 秒)

S6和S7控制信号相同(与S5、S8互补):

$g6=g7$ (方波, 周期 $100\text{e-}6$ 秒, 占空比50%, 相位延迟 $75\text{e-}6$ 秒)

4 转换矩阵Req

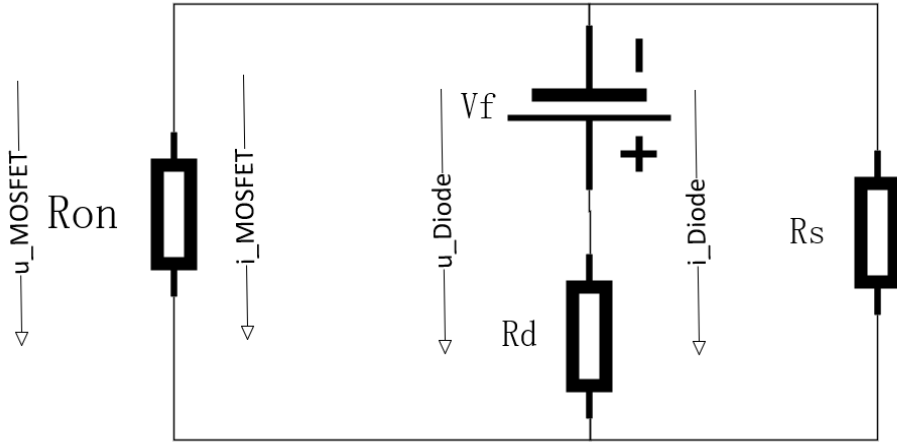


图 2: MOSFET和Diode示意图

MOSFET导通时: $u_{MOSFET} = R_{on} \times i_{MOSFET}$

MOSFET截止时: $i_{MOSFET} = 0$

Diode导通时: $u_{Diode} = R_d \times i_{Diode} - V_f$

Diode截止时: $i_{Diode} = 0$

5 MOSFET和Diode导通状态判断

(g1,g2,g3,g4)控制信号仅有三种情况(1,0,0,1)、(0,1,1,0)、(0,0,0,0)

(g5,g6,g7,g8)控制信号与(g1,g2,g3,g4)处理方法相同。

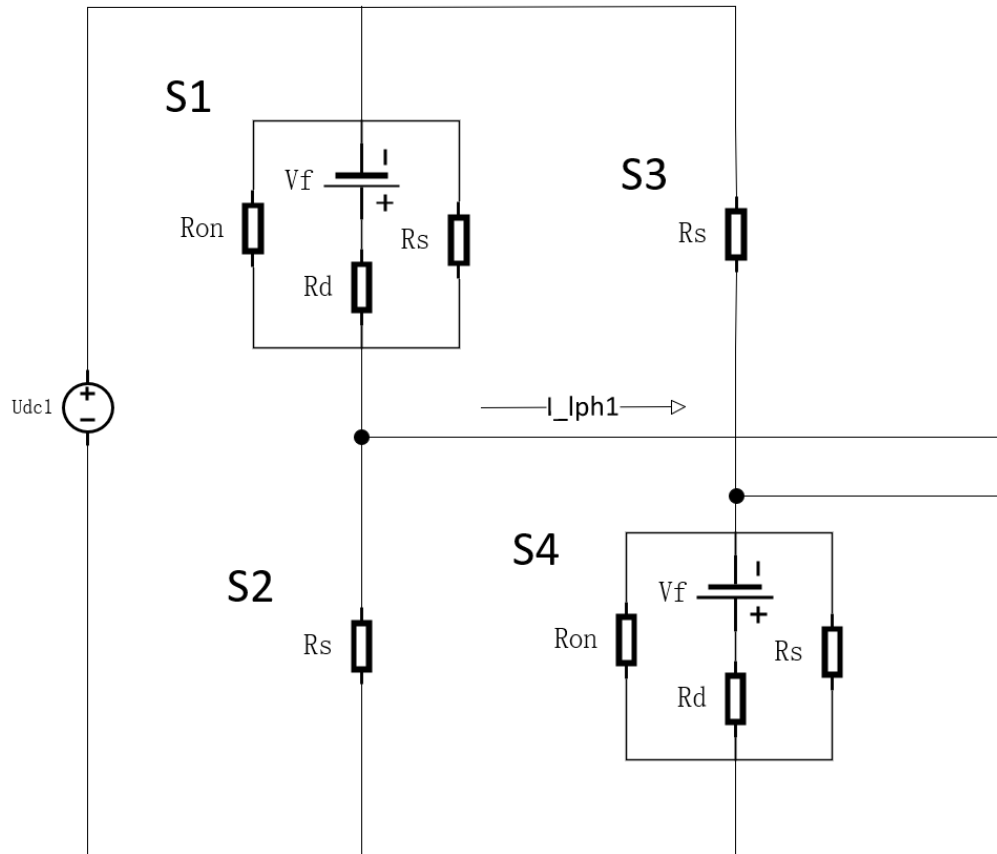


图 3: 控制信号(g1,g2,g3,g4)=(1,0,0,1)情况

S1-MOSFET 导通

S2-MOSFET 截止

S3-MOSFET 截止

S4-MOSFET 导通

S1-Diode 若 I_{lph1} 流出(> 0)则截止, 若 I_{lph1} 流入(< 0)则导通($V_f=0$)

S2-Diode 截止

S3-Diode 截止

S4-Diode 若 I_{lph1} 流出(> 0)则截止, 若 I_{lph1} 流入(< 0)则导通($V_f=0$)

(g1,g2,g3,g4)=(0,0,0,0) 情况仅在开始时可能存在, 桥所有MOSFET和Diode均截止。

6 仿真结果

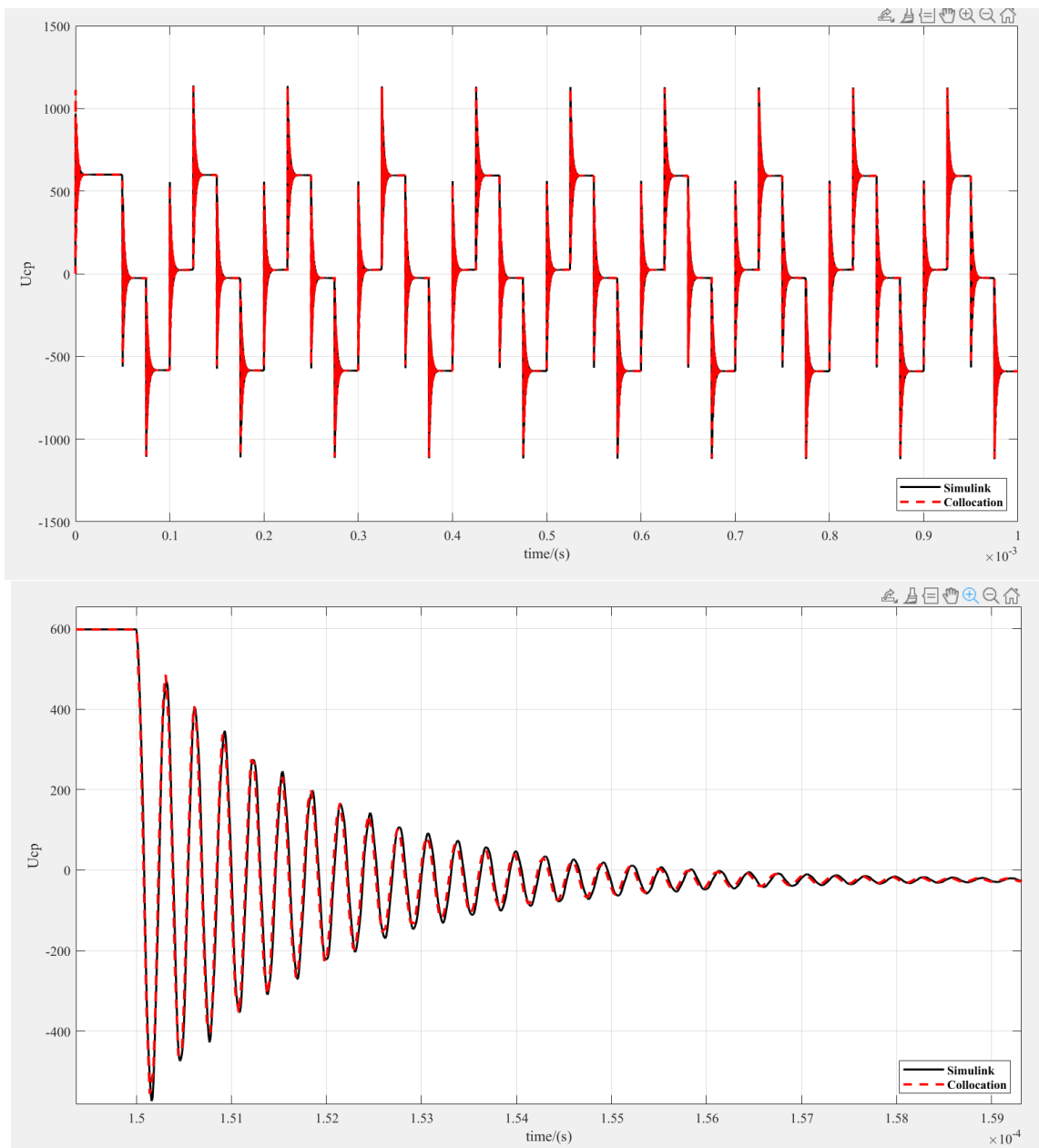


图 4: U_{cp} -时间图像

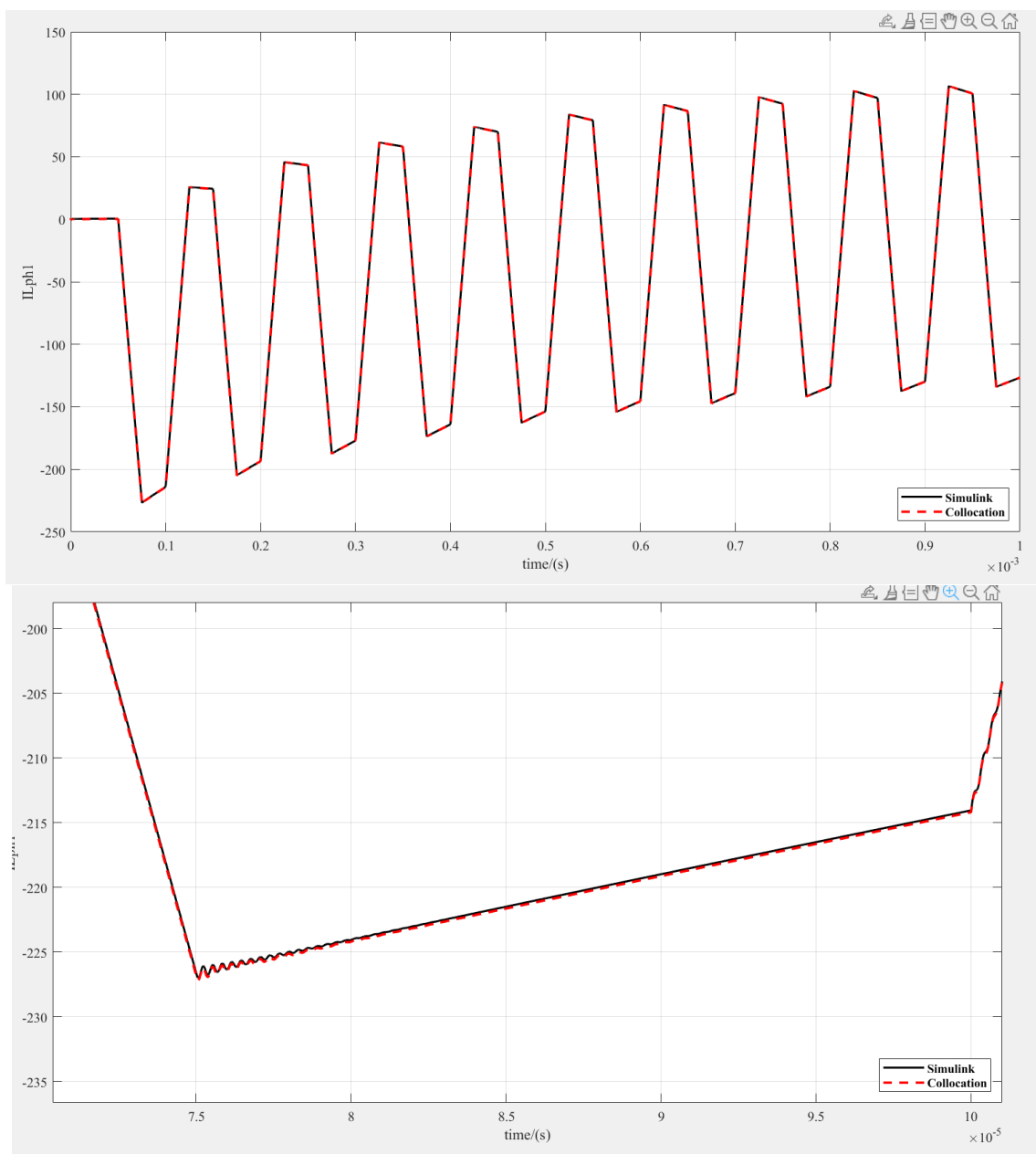


图 5: IL_{ph1} -时间图像